

Response Letter with Annotations

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Annotations

1. Purposes

a. Motivation (one-sentence summary)

Authors were concerned about the effects of social media platforms (Facebook and Instagram) on affective polarization due to the perceived credibility and endorsement of reposting/resharing to users' social media networks, regardless of the content's quality or truthfulness.

b. Specific Rationale (several bullet points)

- Prior research demonstrates that reshared content may increase both political polarization and political knowledge
- Reshares can increase exposure to misinformation and click bait/share bait/rage bait content

c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
- Identify specific relationships among variables

Their main hypotheses were:

H1: Withholding reshared content from users' feeds may reduce affective polarization by decreasing exposure to emotionally or ideologically inflammatory content

H2: Removing reshared content would, on net, reduce accurate knowledge about the election campaign

2. Method

a. Research Design

- Experiment: conditions, manipulations, etc.

The study was an experiment where participants were randomized to either the true control group (no change in feed) and one of the treatments (which removed reshared content from friends, groups or pages, aka the "No Reshares" group) for ~3 months.

b. Sample (representative sample?)

They recruited participants from the entire U.S. Facebook user population through survey popups on the top of Facebook feeds as a part of the larger 2020 Facebook and Instagram Election Study (FIES) which included eight different experimental conditions. People who completed baseline surveys were also invited to share data on both their desktop and mobile web visits as a part of

the web tracking sample. Their final Ns were 23,402 for surveys and 3,781 for the web tracking data.

c. Measures + Analysis

The authors were most interested in the population average treatment effect (PATE) which was weighted by predicted ideology, friend count, number of political pages followed, number of days active, and a series of other variables (in supplemental materials). They also measured various aspects of user behavior on the platform such as time spent, clicks on content, proportion of reactions to views. The authors reported PATEs as their main estimand of interest and also reported sample average treatment effects (SATEs).

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Users in the treatment group spent less time on Facebook and clicked on less content and reacted to proportionally less posts viewed than those in the control group.
- Exposure to content from untrustworthy sources and uncivil posts were relatively low at baseline.
- Feeds without reshares had less content pertaining to politics on average compared to the control group and political news exposure decreased by more than half.
- Proportion of like-minded sources decreased for the treatment group while proportion of feed from more moderate or mixed sources increased

Q1: Support was not found for H1. There were not significantly lower levels of affective or issue polarization in the treatment group compared to the control group.

Q2: Support was not found for H2. There were no significant changes in election knowledge with the PATE. But among the SATE, removing reshares significantly reduced news knowledge overall – with participants less likely to recall recent events in the treatment group.

- They also observed no significant effects for many other comparisons such as perceived accuracy of factual claims, trust in media, confidence in political institutions, perceptions of polarization, belief in legitimacy of the election, support for political violence, among other outcomes.
- The treatment group clicked significantly less frequently on political news content from partisan sources.
- Participants in the no reshares group had significantly lower news knowledge, and were less likely to be able to discern false claims after multiple comparison adjustments. But these were unregistered analyses conducted by the authors of their own accord and interest.

b. Include Key Figures/Tables

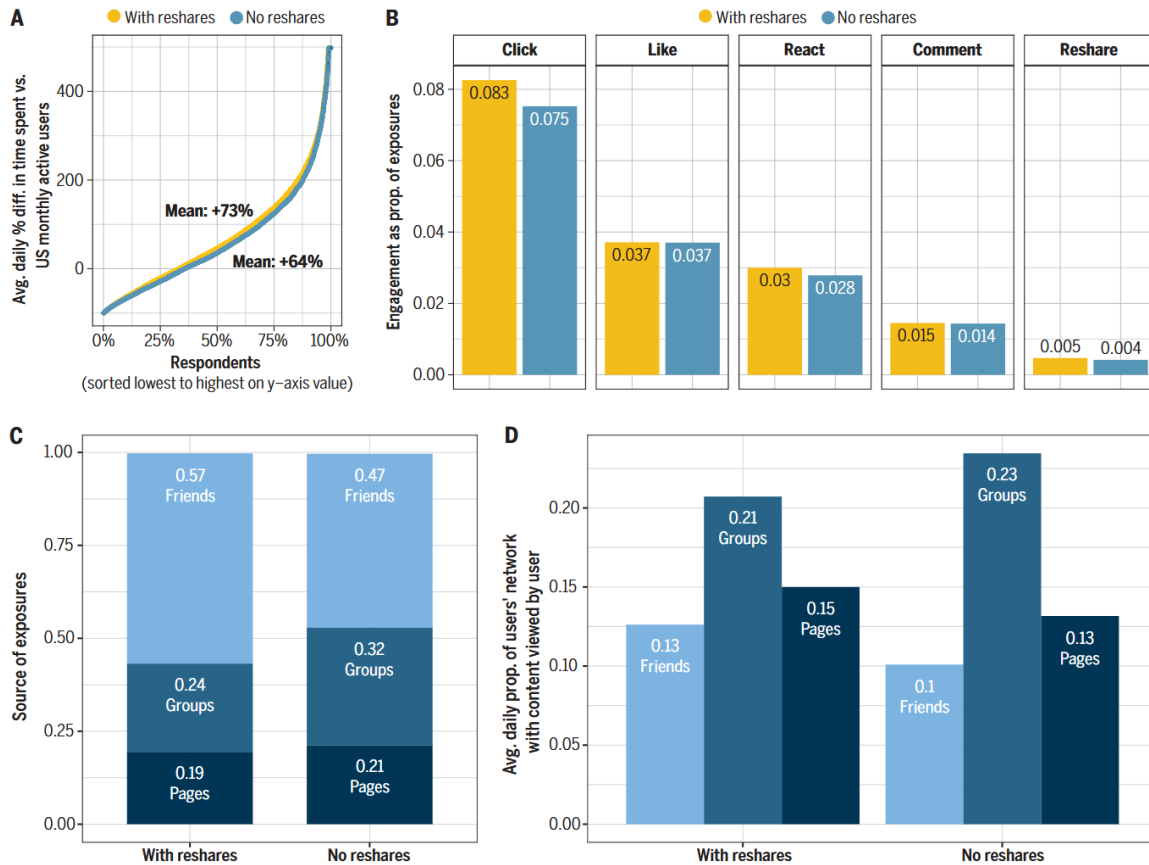


Fig. 1. Comparison of user experience and behavior for the No Reshares treatment and control conditions. (A to D) Values are unweighted sample statistics ($N = 23,402$). All differences are significant at the $p < 0.005$ level except for panel B Click rate ($p < 0.05$), Like rate ($p > 0.05$), Comment rate ($p > 0.05$), and Reshare rate ($p > 0.05$); confidence intervals are thus not shown.

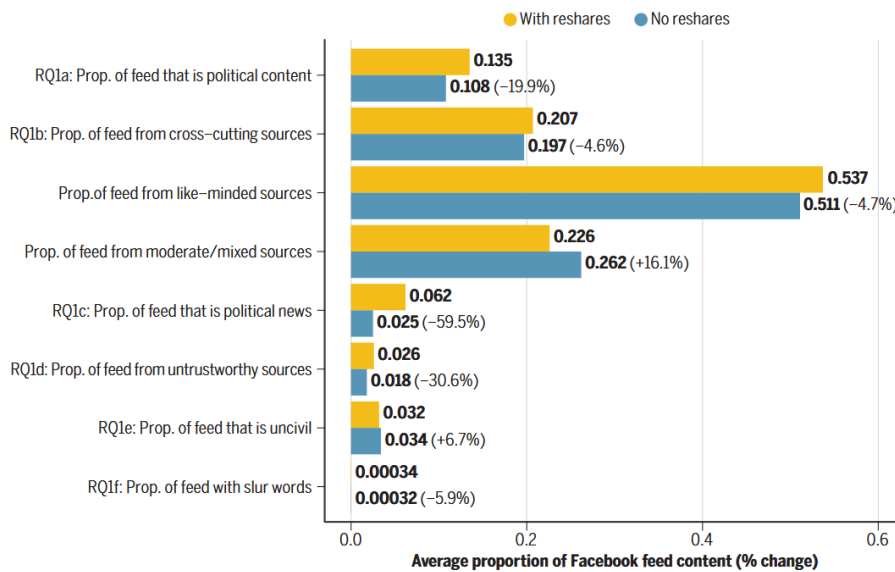


Fig. 2. Estimated changes in prevalence of feed content on Facebook. Values are average unweighted proportions within each group, with percent changes relative to the control group in parentheses ($N = 23,402$). All differences are significant at the $p < 0.005$ level, except RQ1f ($p > 0.05$); confidence intervals are thus not shown. Fully specified regression models with survey weights are reported in SM section S22.

Table 1. Population average treatment effects (PATE) and sample average treatment effects (SATE) of the No Reshares group (relative to the control group) for primary hypotheses (top) and secondary hypotheses (bottom). Estimates are presented in standard deviations (SD), and *p*-values are presented with and without sharpened FDR adjustment (26). Outcomes marked with * were computed using passive tracking data subsample (*N* = 3,781; see SM section S9). Outcomes marked with † were computed using Facebook platform data (*N* = 23,402; see SM section S6). All other outcomes were computed using survey data (*N* = 23,402; see SM section S1.5).

Hypothesis	Estimand	ATE	95% CI	SE	<i>p</i>	<i>P</i> _{adj}
H1a: Affective polarization	PATE	−0.006	[−0.034, 0.022]	0.014	0.673	1.000
	SATE	0.012	[−0.005, 0.030]	0.009	0.158	0.188
H1b: Issue polarization	PATE	0.012	[−0.018, 0.042]	0.015	0.418	0.895
	SATE	0.007	[−0.010, 0.025]	0.009	0.414	0.268
H2a: Election knowledge	PATE	−0.017	[−0.064, 0.030]	0.024	0.472	0.895
	SATE	−0.021	[−0.048, 0.006]	0.014	0.124	0.188
H2b: News knowledge	PATE	−0.053	[−0.101, −0.004]	0.025	0.034	0.157
	SATE	−0.069	[−0.095, −0.043]	0.013	<0.01	<0.01
SH1a: Factual discernment	PATE	−0.045	[−0.082, −0.008]	0.019	0.016	0.151
	SATE	−0.027	[−0.048, −0.005]	0.011	0.015	0.083
SH1b: Off-platform political news visits*	PATE	0.061	[−0.042, 0.164]	0.052	0.243	0.751
	SATE	0.044	[−0.022, 0.111]	0.034	0.193	0.348
SH2a: Trust in media (excluding social)	PATE	−0.015	[−0.042, 0.012]	0.014	0.264	0.751
	SATE	−0.017	[−0.033, −0.002]	0.008	0.031	0.122
SH2b: Trust in info from social media	PATE	0.017	[−0.020, 0.053]	0.019	0.371	0.782
	SATE	−0.013	[−0.035, 0.009]	0.011	0.245	0.416
SH2c: Confidence in institutions	PATE	0.021	[−0.024, 0.067]	0.023	0.354	0.782
	SATE	0.010	[−0.015, 0.035]	0.013	0.445	0.506
SH3a: Perceived polarization	PATE	−0.028	[−0.066, 0.011]	0.020	0.157	0.579
	SATE	−0.008	[−0.029, 0.014]	0.011	0.480	0.506
SH3b: Partisan news clicks†	PATE	−0.109	[−0.159, −0.059]	0.025	<0.01	<0.01
	SATE	−0.105	[−0.128, −0.082]	0.012	<0.01	<0.01
SH3c: Partisan news visits*	PATE	−0.049	[−0.152, 0.054]	0.052	0.348	0.782
	SATE	0.004	[−0.066, 0.074]	0.036	0.900	0.713
SH4: Epistemic political efficacy	PATE	0.032	[−0.006, 0.070]	0.019	0.098	0.521
	SATE	0.017	[−0.006, 0.039]	0.011	0.146	0.311
SH5a: Party-line presidential voting	PATE	0.002	[−0.033, 0.037]	0.018	0.920	1.000
	SATE	−0.004	[−0.022, 0.014]	0.009	0.655	0.575
SH5b: Party-line down-ballot voting	PATE	0.012	[−0.036, 0.060]	0.025	0.625	0.966
	SATE	0.012	[−0.016, 0.040]	0.014	0.391	0.506
SH6: Belief in legitimacy of the election	PATE	−0.026	[−0.061, 0.009]	0.018	0.143	0.579
	SATE	−0.019	[−0.038, 0.001]	0.010	0.058	0.176
SH7: Support for political violence	PATE	−0.008	[−0.051, 0.035]	0.022	0.718	0.976
	SATE	−0.003	[−0.029, 0.024]	0.014	0.840	0.713

c. Authors' Conclusions

The authors concluded that reshares are a “double edged sword” where reshares inform users about politics and current events but do so with less trustworthy sources that exaggerate (baity articles) or purely fabricate information. They acknowledge they were unable to determine any impacts on polarization or shifts in political attitudes and behaviors, but emphasize that partisan clicks did reduce among the no reshares treatment group. They overall state that Facebook reshares are likely not the main factor of influencing user attention and behavior during the 2020 election campaign.

4. Discussion

a. Implications

b. Limitations

The authors were limited by their study design because they couldn't estimate some kind of "societal impact" about what drives engagement or "demand for certain kinds of content" or the incentives for making popular content. They also note that their study's length was in the short term over the course of a few months, when they are entering late in the game as observable polarization and divisiveness in the U.S. was already high.

Response Letter (~600 words)

- Your assessment of the study

I think this study was surprisingly small in scale (as far as sample size) despite the multimillions of people who use the platform (though survey research always struggles with response rates). That being said, their study was powered enough to be able to calculate PATEs which isn't all that common even among my health services research knowledge. I think the study is a reasonable extension of past academic literature to apply to the concerns that brought about the Facebook & Instagram Election Study.

- Weaknesses

A handful of things about this study were unclear to me. The authors wrote about website and mobile site visits, but I don't think they explicitly stated that app use was included (but maybe that was considered part of mobile visits). I also think the authors were correct that they are entering "too late into the game" to see any shifts on polarization – the polarizing already happened prior to the study gradually over time (years, if I had to guess).

Considering ethics, I'm also not sure if Facebook was actually as open to investigation as *they wanted to appear to be*. Yes, they allowed use of the site to recruit participants, yes, they allowed independent researchers to conduct surveys and experiments and access certain kinds of data related to the platform's use *during the study's duration*. But I'm unsure if looking for effects of polarization through one of the metrics that drive engagement (reshares) a month prior to the election and a bit after, actually tells us anything about how Facebook may have manipulated algorithmically the kinds of content that users would see over time. I would be more interested if they were able to use some sort of quasi-experimental method with retrospective data about the platform's engagement over multiple years leading up to the election. But I suppose that would be out of scope for what this team in particular was interested in. One remaining observation that I won't give much weight – but the study's details feel lukewarm findings-wise, but certainly has a flashy title that makes Facebook look innocuous.